Huawei Access Node MG-ACC-46 MG Road Core QoS Congestion Event ALARM_8804 SOP

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8804

Description

This SOP addresses the occurrence of QoS congestion events on the Huawei Access Node MG-ACC-46 located at the MG Road Core site. The failure domain in question pertains to microbursts, interface tail drops, and QoS queue starvation, as indicated by the ALARM_8804 severity of Warning. The root cause of this issue can be attributed to the inherent limitations of the Huawei access node's QoS mechanism, which relies on the DiffServ protocol to manage packet transmission. The DiffServ architecture allocates traffic classes to packets based on their IP header information, assigning precedence levels that dictate queuing and shaping behavior. However, in scenarios where microbursts occur, the QoS engine may struggle to adapt, leading to interface tail drops and subsequent QoS queue starvation. Furthermore, the Huawei access node's limited buffer size may exacerbate this issue, as packets are dropped before they reach the QoS engine, triggering the ALARM_8804 warning.

Pre-requisite Checks

- Verify that the MG Road Core site is accessible and that remote hands are dispatched for any necessary physical interventions.

- Check the status of the redundant link to ensure it is operational and not contributing to the congestion event.

- Verify that the Huawei access node MG-ACC-46 is powered on and that all interfaces are functioning as expected.

Potential Root Causes

Cause	Probability
Microburst-induced QoS queue starvation due to insufficient DiffServ buffer size.	High
Interface tail drop due to oversubscription and inadequate buffer allocation.	Medium
QoS engine misconfiguration or inadequate policing mechanism.	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue, Expected Output: Queue information including QoS queue status and packet drop rates., Description: This command is used to diagnose QoS queue performance and identify potential causes of congestion events.

Command: display interface brief, Expected Output: Interface status including bandwidth utilization and packet drop rates., Description: This command is used to diagnose interface performance and identify potential causes of congestion events.
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```
Command: display qos traffic-statistics, Expected Output: QoS traffic statistics including packet counts and drop rates., Description: This command is used to diagnose QoS traffic performance and identify potential causes of congestion events.
```

Resolution Steps

Step 1: Step: Modify DiffServ buffer size to increase queue capacity., Command: qos queue modify , Description: This step involves increasing the DiffServ buffer size to accommodate more packets in the queue, thereby reducing the likelihood of QoS queue starvation.

Step 2: Step: Adjust interface configuration to optimize bandwidth allocation., Command: interface bandwidth , Description: This step involves adjusting the interface configuration to optimize bandwidth allocation and reduce oversubscription.

Step 3: Step: Reconfigure QoS engine to improve policing mechanism., Command: qos engine reconfigure , Description: This step involves reconfiguring the QoS engine to improve the policing mechanism and prevent QoS queue starvation.

Post-Check Verification

```
Command: display qos queue, Expected Output: Queue information indicating improved QoS queue performance and reduced packet drop rates., Description: This command is used to verify the effectiveness of the resolution steps and ensure that QoS queue performance has improved.

Command: display interface brief, Expected Output: Interface status indicating improved interface performance and reduced packet drop rates., Description: This command is used to verify the effectiveness of the resolution steps and ensure that interface performance has improved.

Command: display qos traffic-statistics, Expected Output: QoS traffic statistics indicating improved QoS traffic performance and reduced packet drop rates., Description: This command is used to verify the effectiveness of the resolution steps and ensure that QoS traffic performance has improved.
```

Roll-back Procedures

- Step: Restore original DiffServ buffer size., Command: qos queue restore , Description: This step involves restoring the original DiffServ buffer size in case the resolution steps fail to resolve the issue.

- Step: Revert interface configuration to original settings., Command: interface revert, Description: This step involves reverting the interface configuration to its original settings in case the resolution steps fail to resolve the issue.

- Step: Revert QoS engine configuration to original settings., Command: qos engine revert , Description: This step involves reverting the QoS engine configuration to its original settings in case the resolution steps fail to resolve the issue.